

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Data Structures & Algorithms, Distributed Systems, Database Systems, Operating Systems, Computer Networks
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf for schema evolution and cross-team compatibility at global scale
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven TypeScript/React system with observer pattern across **25K+ lines** of Chromium at **95% test coverage**; design documents adopted into production branch by senior Chrome engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Designed **idempotent payment APIs** with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced **exactly-once semantics** preventing duplicate transactions and data inconsistency under network partitions
- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key **schema** — **30% read throughput improvement** under sustained high-concurrency traffic

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned PostgreSQL **schema** and indexing strategy achieving **sub-100ms** response for 10,000+ records

PERSONAL PROJECTS

TiMoto AI — Consumer AI Platform | *Go, vLLM, Django, PostgreSQL, AWS, Terraform*

timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end consumer AI evaluation platform: inference engine, gRPC inter-service layer, Django REST backend, PostgreSQL, and multi-AZ AWS cloud infrastructure
- Resolved **production deadlock under concurrent requests** — traced root cause to circular resource acquisition; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with Terraform IaC and circuit breaker pattern — **99.9% uptime, 44% cost reduction** to \$40–60/month; auto-rollback on health check failure

Pulumi — Open Source Contributor | *Go, TypeScript, IaC*

24.4K★ github.com/pulumi/pulumi

- Submitted **Go CLI** features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Languages: Go, C++, Python, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Frameworks & Databases: Node.js, React, Django, FastAPI, REST APIs, PostgreSQL, Redis, MongoDB

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Idempotent APIs, Event-driven architecture, Fault tolerance

Performance: p99 latency profiling, Lock-free data structures, Hot-partition diagnosis, Schema optimization, Sub-50ms SLOs